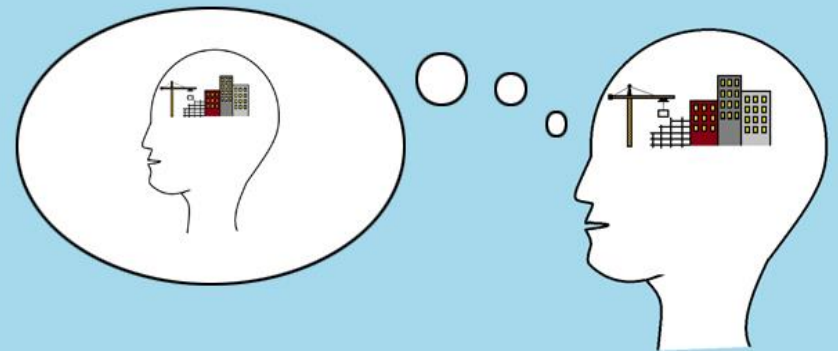


How or why did my intervention work?

Challenges in evaluating mechanisms and program theories



[@heinonmatti](https://twitter.com/heinonmatti)

www.mattiheino.com

What do you mean “it worked”?

- A predicted change in a pre-specified outcome:
Physical activity

Hankonen et al. *BMC Public Health* (2016) 16:451
DOI 10.1186/s12889-016-3094-x

BMC Public Health

STUDY PROTOCOL **Open Access**

 CrossMark

‘Let’s Move It’ – a school-based multilevel intervention to increase physical activity and reduce sedentary behaviour among older adolescents in vocational secondary schools: a study protocol for a cluster-randomised trial

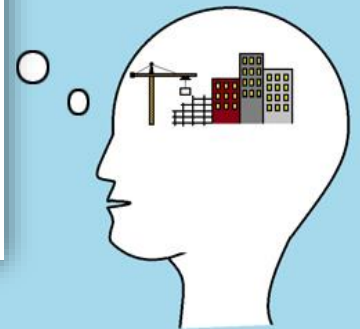
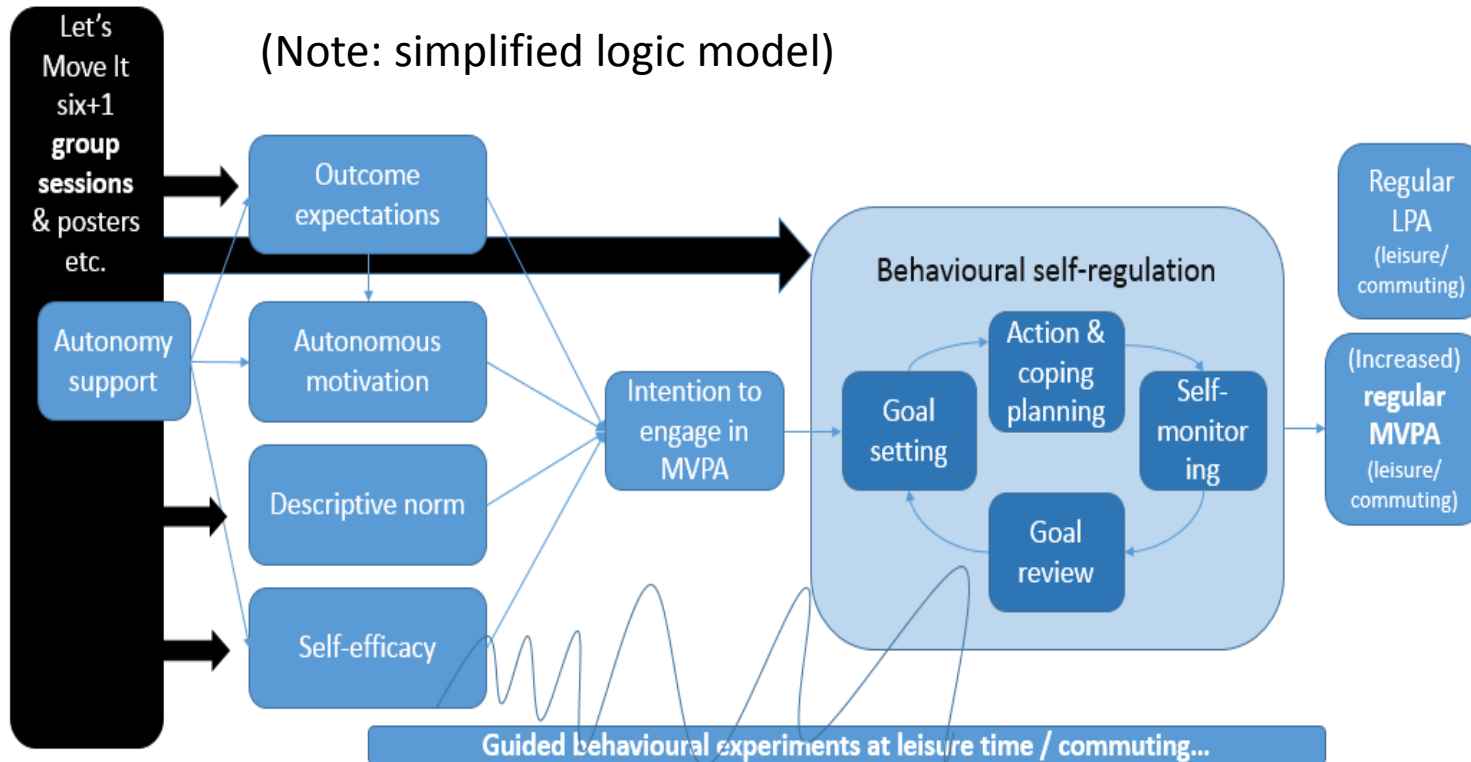
Nelli Hankonen^{1,2*}, Matti T. J. Heino², Vera Araujo-Soares³, Falko F. Sniehotta³, Reijo Sund², Tommi Vasankari⁴, Pilvikki Absetz⁵, Katja Borodulin⁶, Antti Uutela⁶, Taru Lintunen⁷ and Ari Haukkala²

Abstract

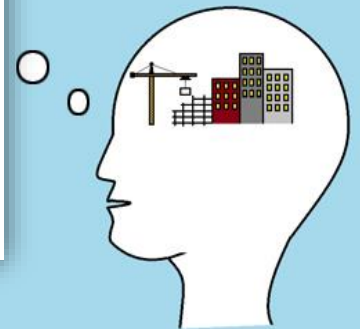
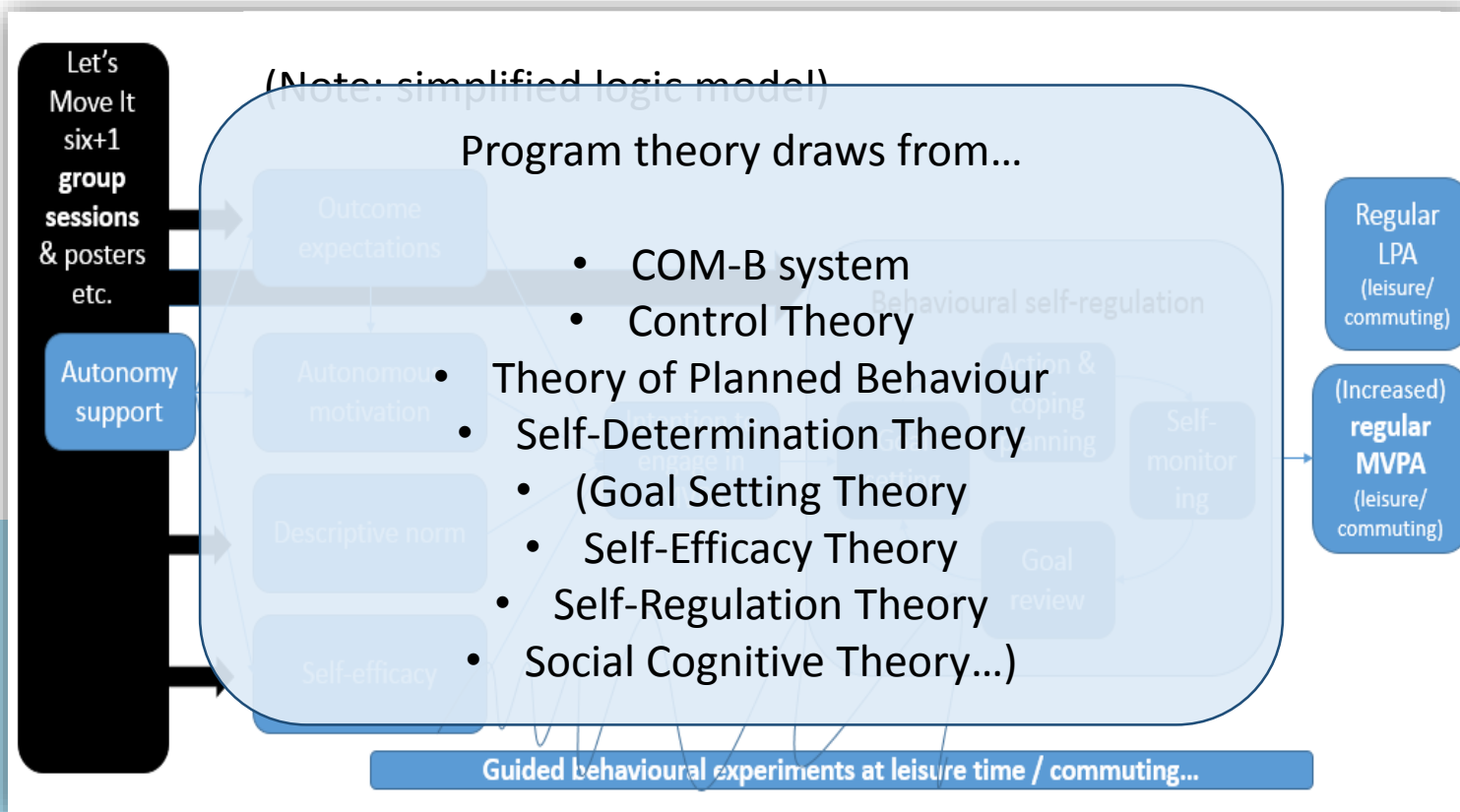
Background: Physical activity (PA) has been shown to decline during adolescence, and those with lower education have lower levels of activity already at this age, calling for targeted efforts for them. No previous study has demonstrated

What do you mean “this is how it worked”?

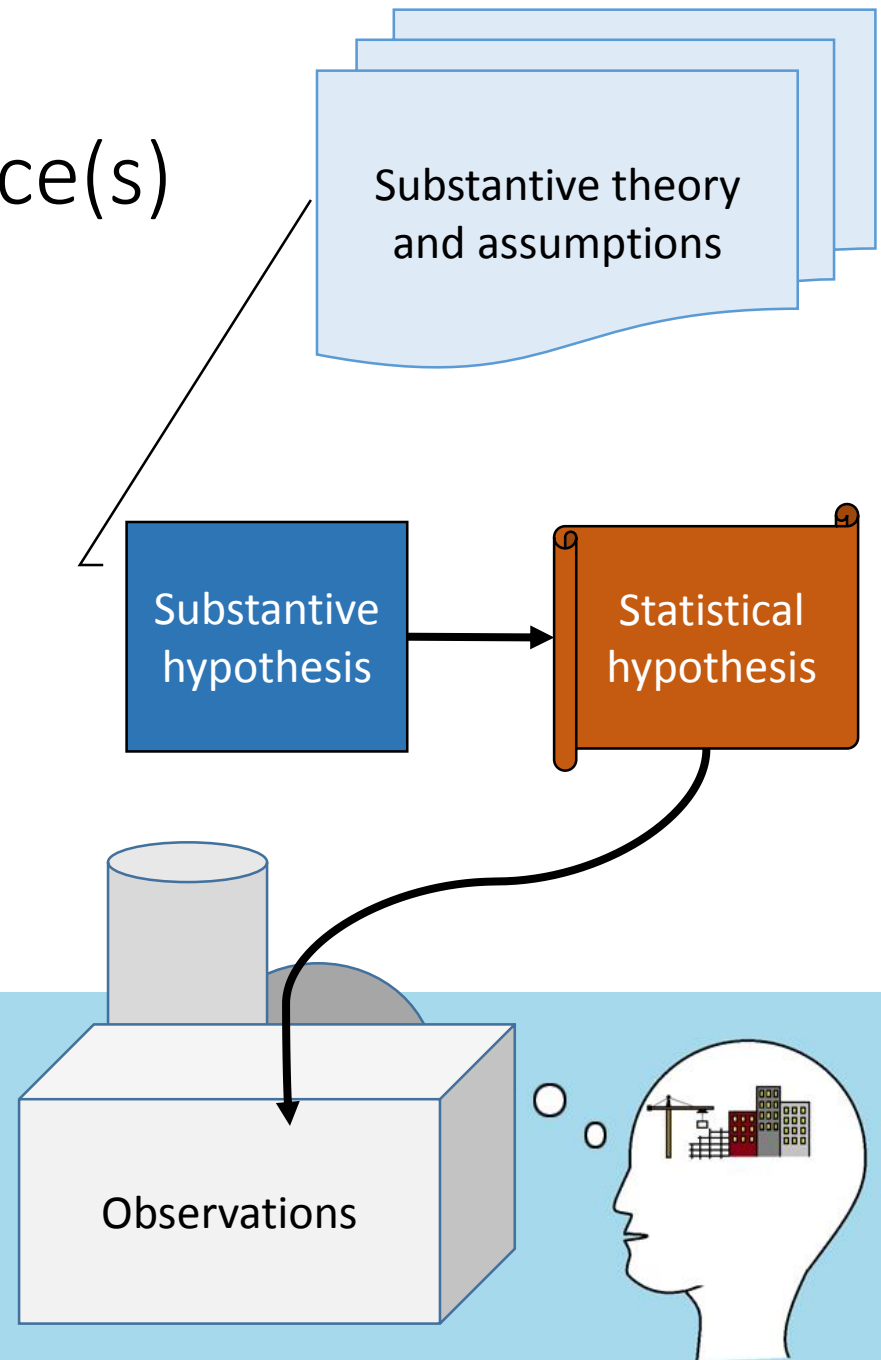
(Note: simplified logic model)



What do you mean “this is how it worked”?

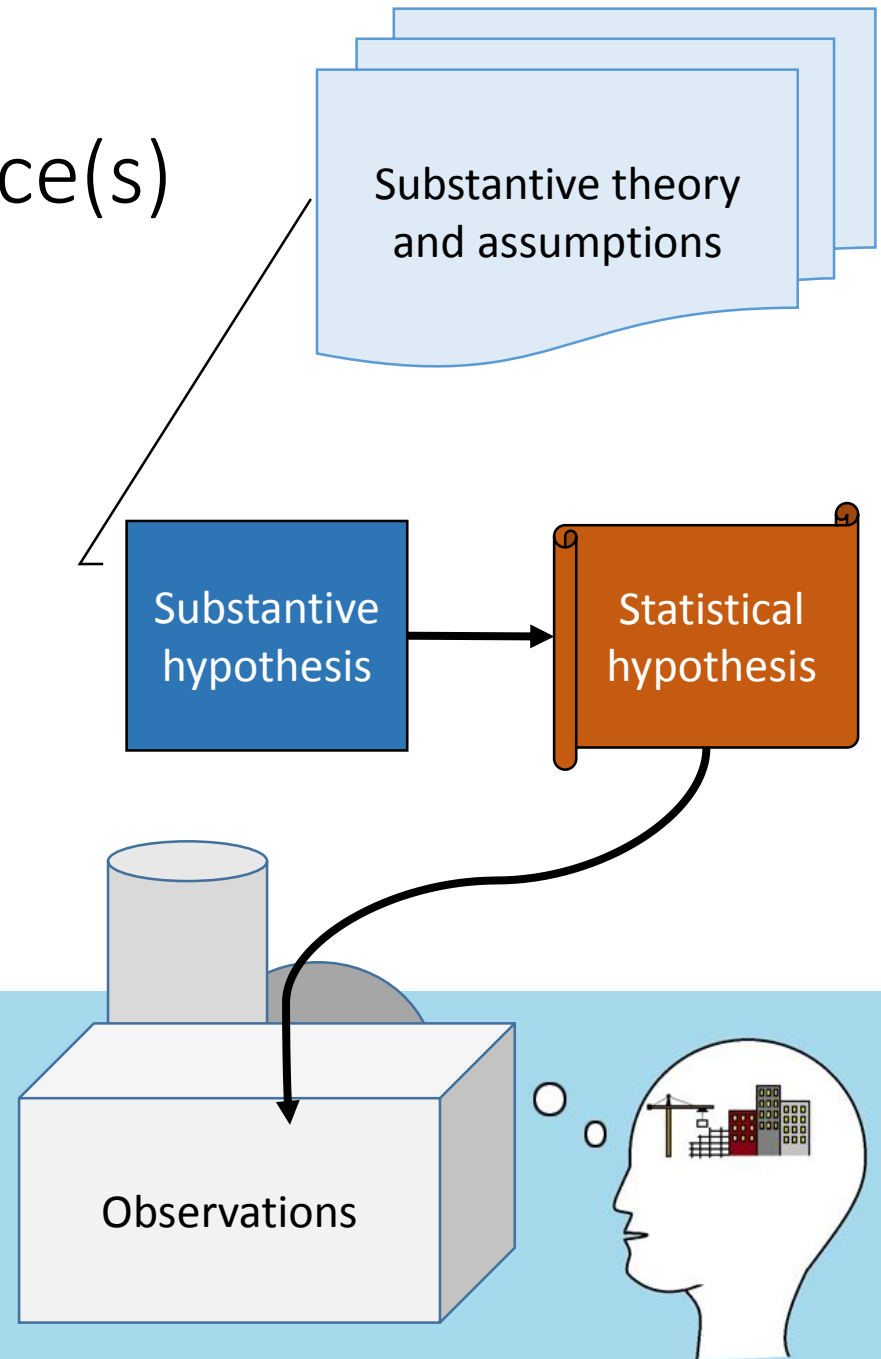


Theory testing in science(s)



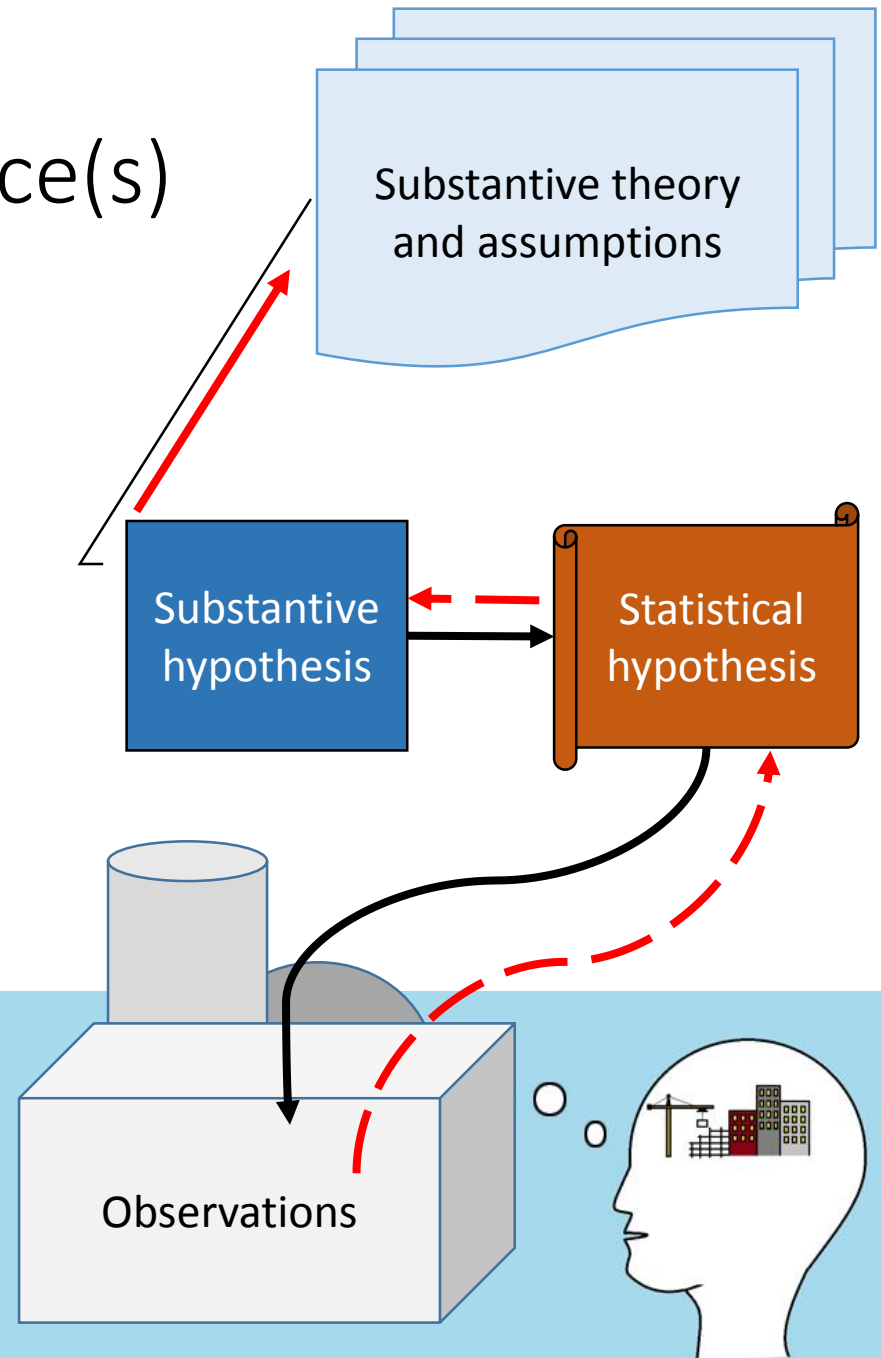
Theory testing in science(s)

↓ The causal stream: each line means “From this, it follows that...” – deductive!



Theory testing in science(s)

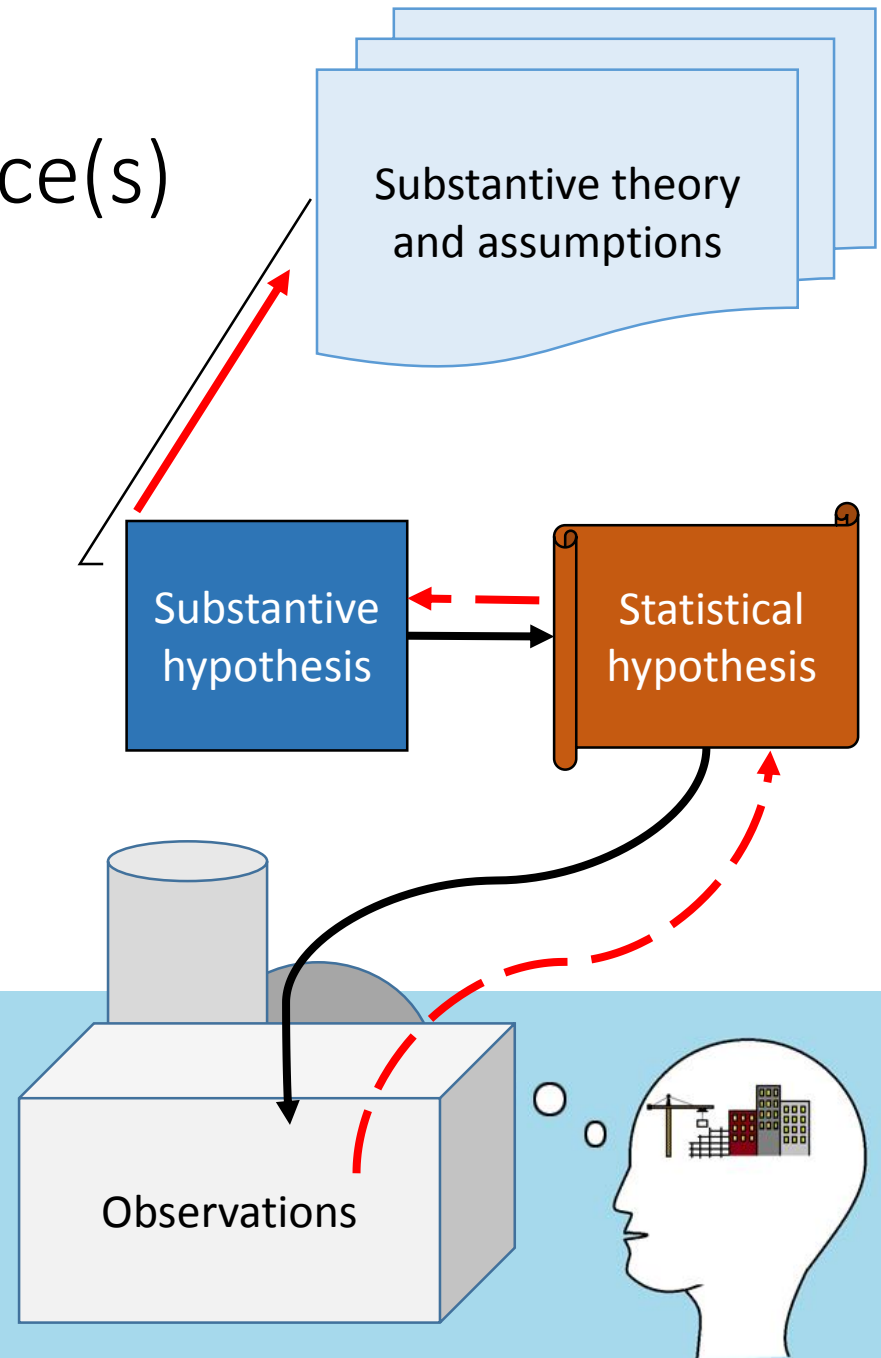
↓ The causal stream: each line means “From this, it follows that...” – deductive!



Theory testing in science(s)

↓ The causal stream: each line means “From this, it follows that...” – deductive!

↑ The inferential stream: each line means “From this, what can we conclude about...?” – inductive!



Theory testing in science(s)

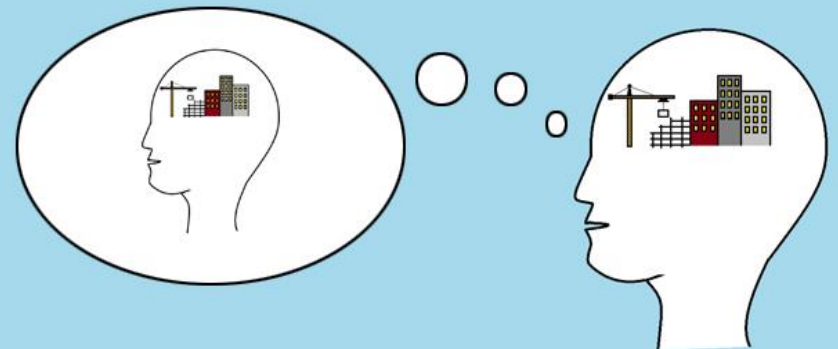
Theory

Auxiliary
theories

All else equal
(*ceteris paribus*)

Assumptions of
instruments

Experimental
conditions



Theory

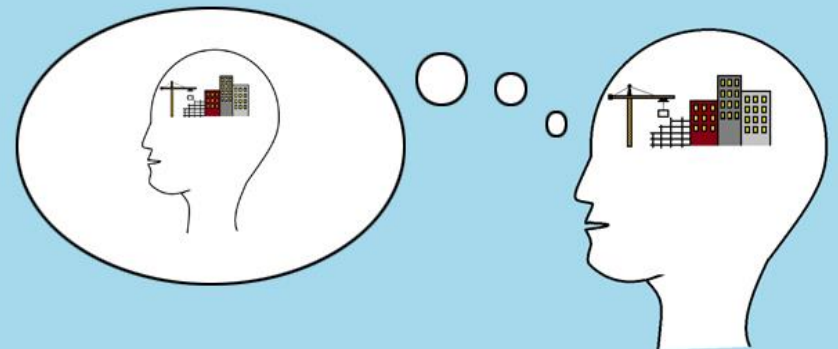
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Program
theory holds



Theory

Auxiliary
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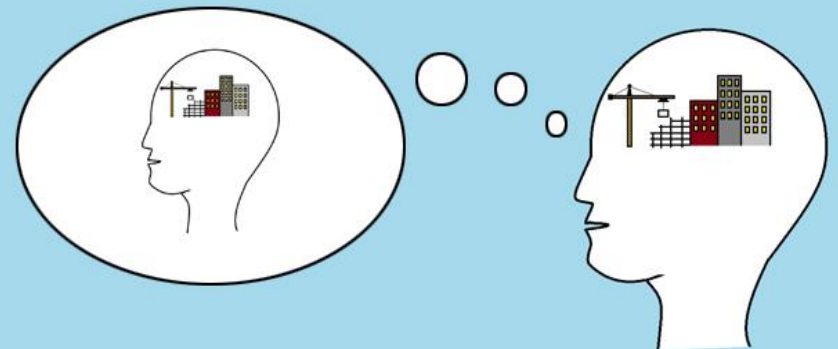
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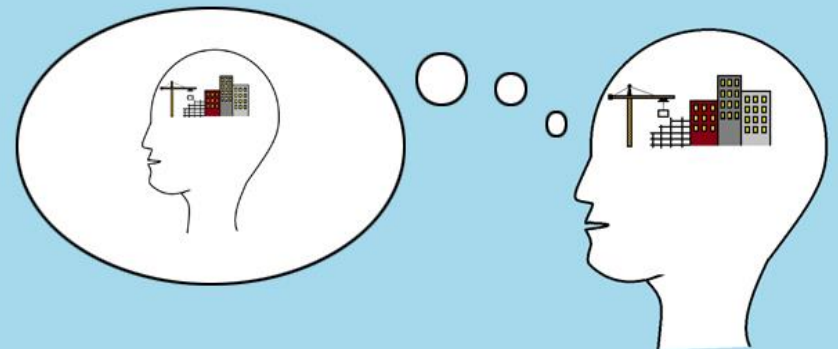
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No systematic,
unmeasured
confounders



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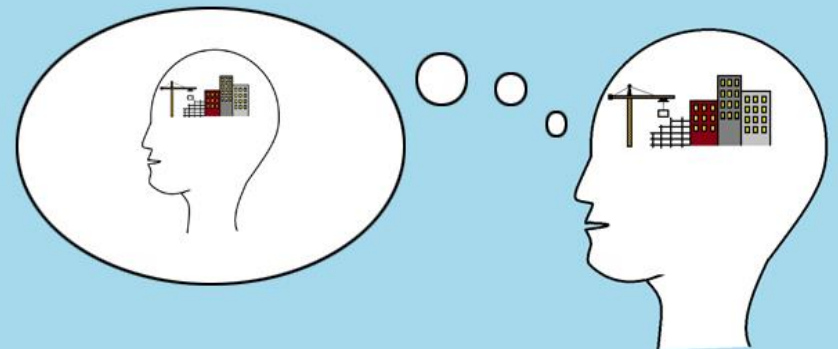
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Assumptions of
instruments

Accelerometers,
bioimpedance
devices etc.
measure what
they're
supposed to

Experimental
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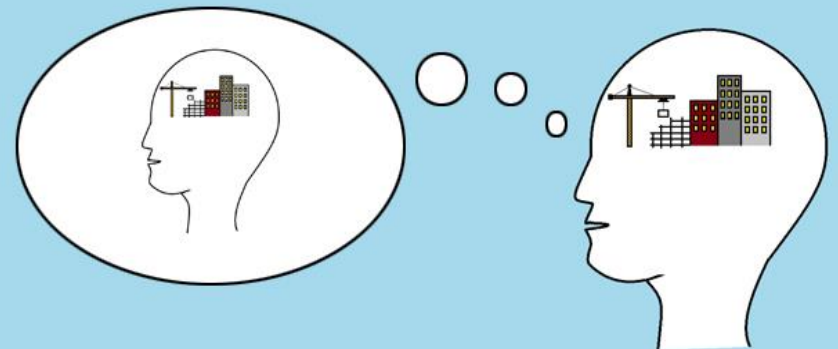
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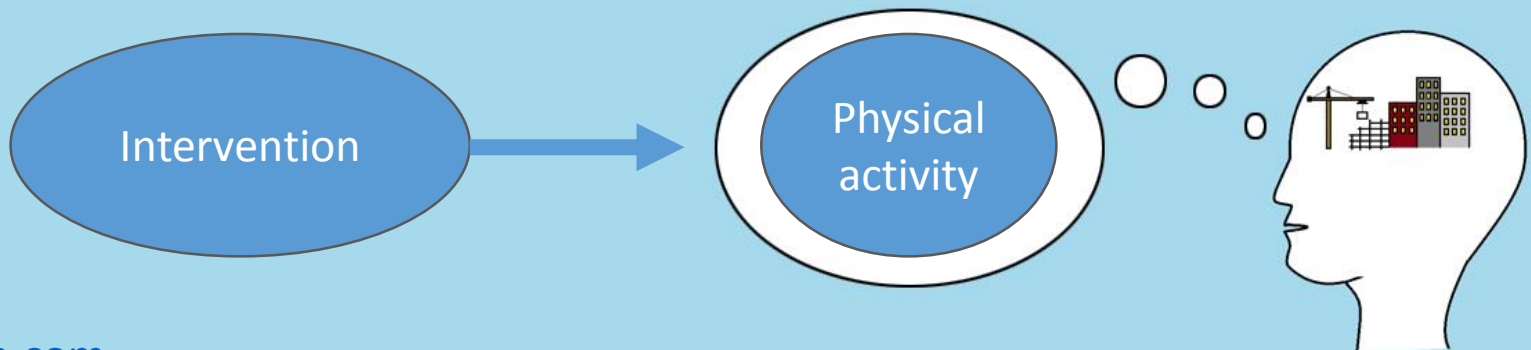
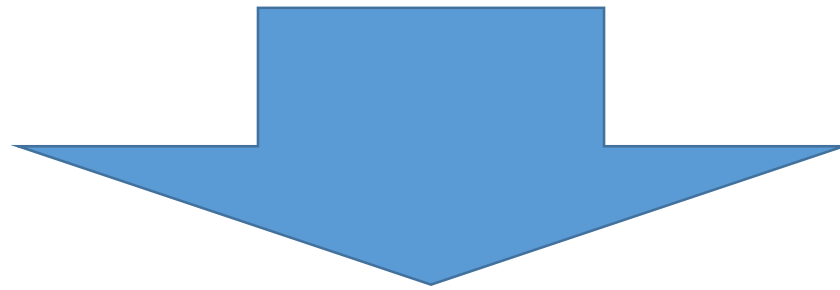
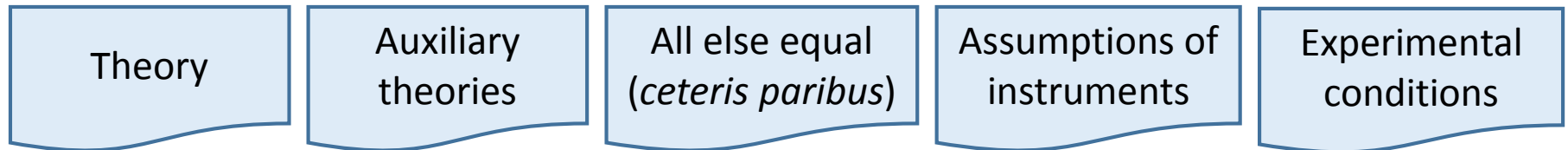
Accelerometers,
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measure what
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Experimental
conditions

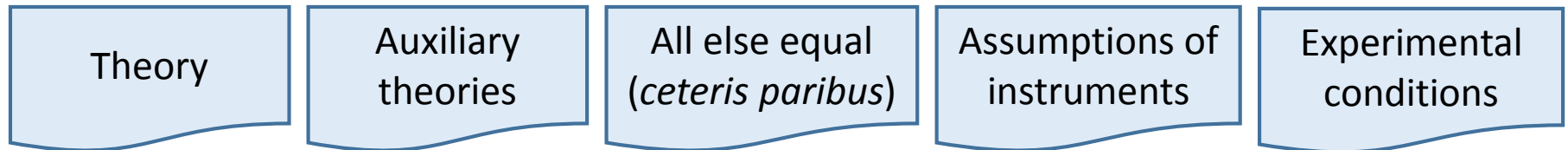
Intervention
delivered as
planned



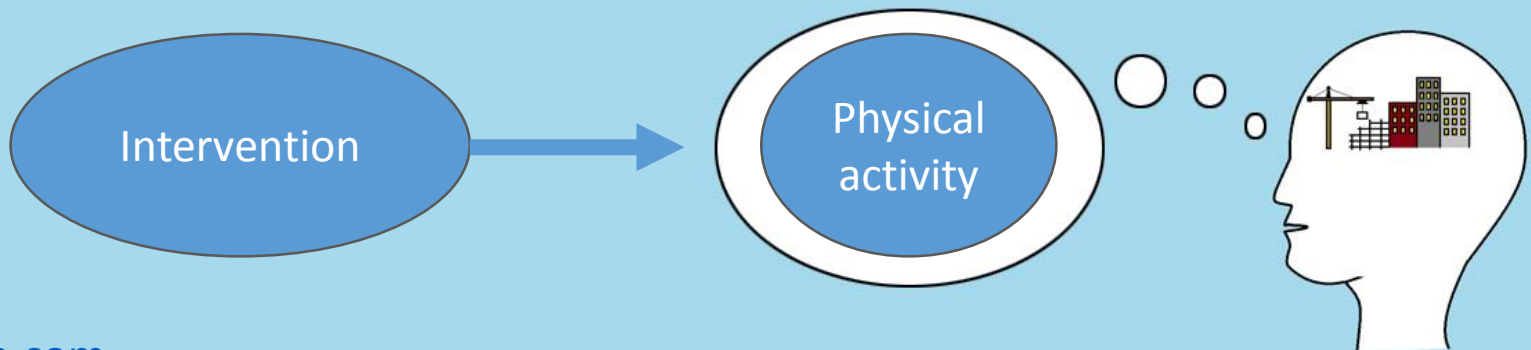
Intervention → PA link tells us something's going on



Intervention → PA link tells us something's going on



... But only to the extent, that increased PA (in comparison to controls) was a “damn strange coincidence” absent intervention.
(see [Meehl 1990](#))



Intervention → PA link tells us something's going on

Theory

Auxiliary
theories

All else equal
(*ceteris paribus*)

Assumptions of
instruments

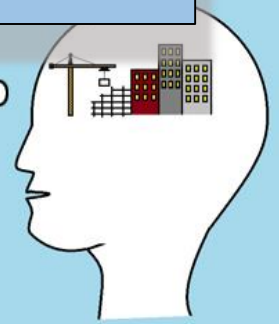
Experimental
conditions

“the only admissible positive evidence for a theory are the corpses of its rivals.”

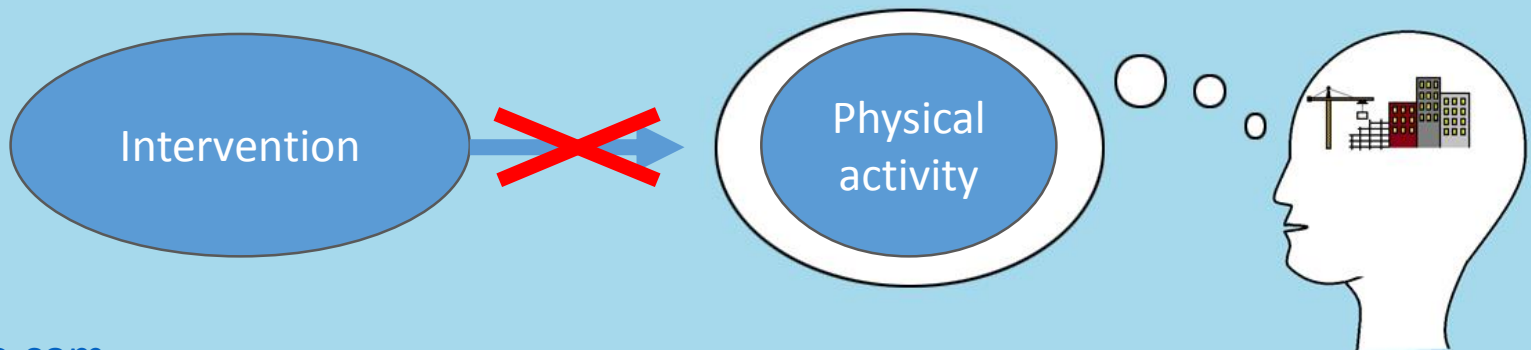
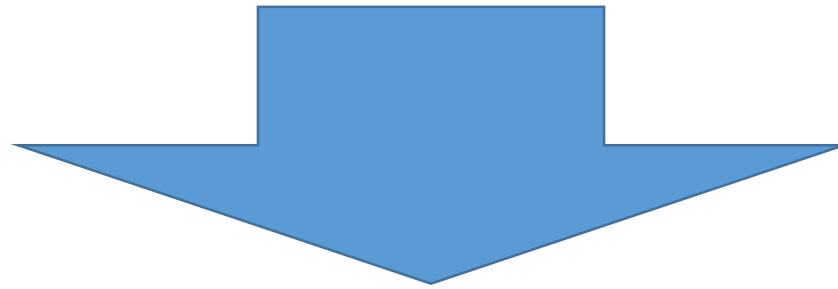
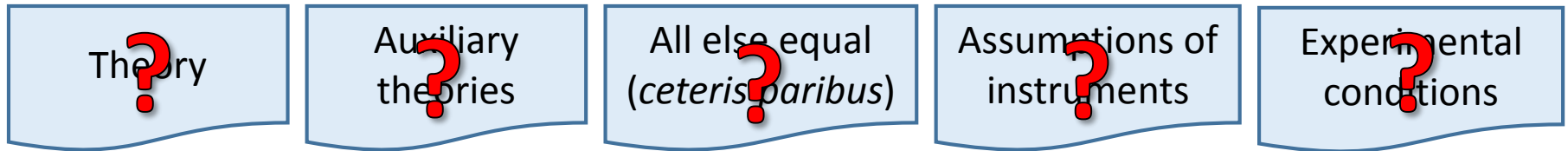
– Lakatos ([1978](#), on Popper's solution)

Intervention

Physical
activity



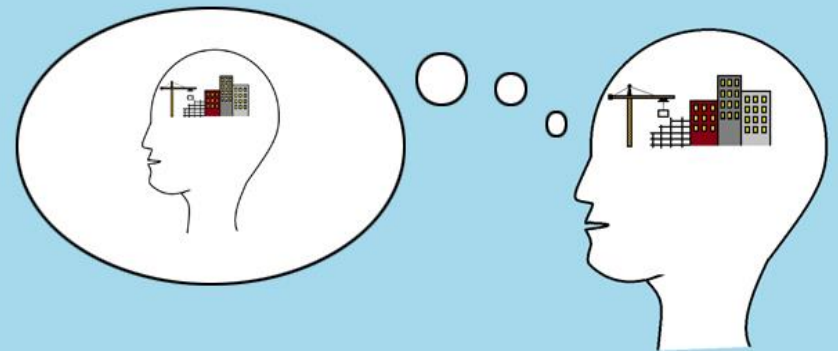
Absence of link only tells us *something* was wrong!



Theory testing in science(s)

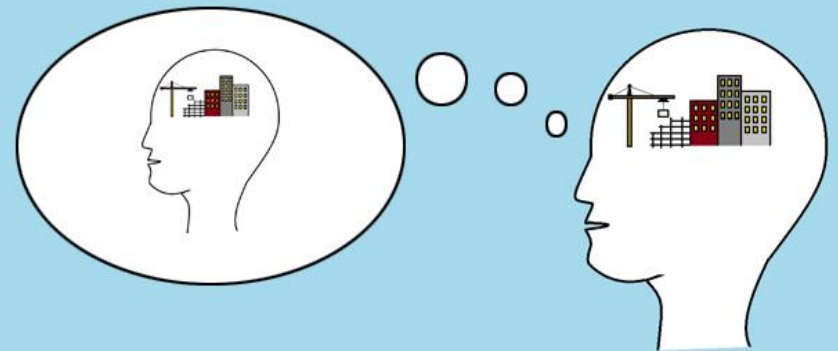
1. Replicate experiment to see that finding holds

- “we may say that a phenomenon is experimentally demonstrable when we know how to conduct an experiment which will rarely fail to give us a statistically significant result.”
 - Fisher (1947); see [Mayo 2016](#)
- Also, see Heino, Fried & LeBel ([2017](#))



Theory testing in science(s)

1. Replicate experiment to see that finding holds
2. Investigate boundary conditions and mechanisms of action



About those mechanisms...

Journal of Personality and Social Psychology
2010, Vol. 98, No. 4, 550–558

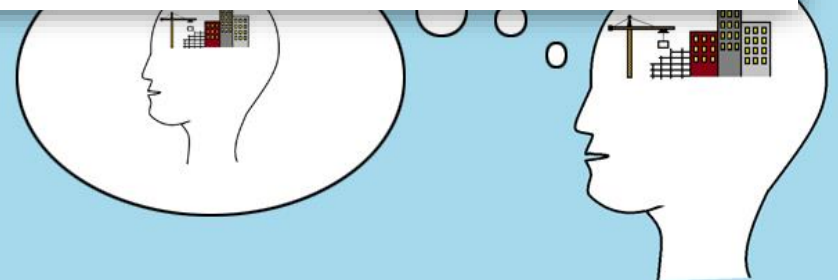
© 2010 American Psychological Association
0022-3514/10/\$12.00 DOI: 10.1037/a0018933

Yes, But What's the Mechanism? (Don't Expect an Easy Answer)

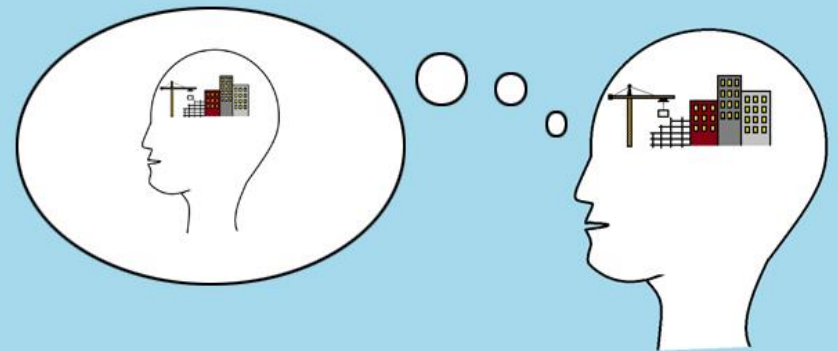
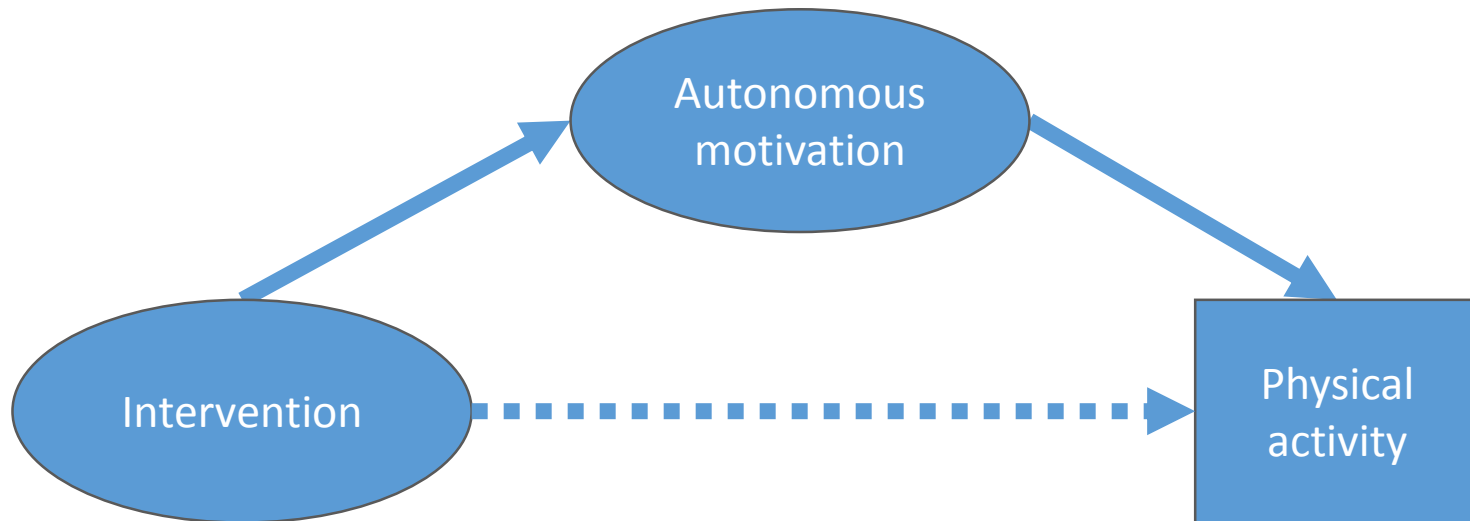
John G. Bullock and Donald P. Green
Yale University

Shang E. Ha
Brooklyn College of the City University of New York

Psychologists increasingly recommend experimental analysis of mediation. This is a step in the right direction because mediation analyses based on nonexperimental data are likely to be biased and because experiments, in principle, provide a sound basis for causal inference. But even experiments cannot overcome certain threats to inference that arise chiefly or exclusively in the context of mediation analysis—threats that have received little attention in psychology. The authors describe 3 of these threats and suggest ways to improve the exposition and design of mediation tests. Their conclusion is that inference about mediators is far more difficult than previous research suggests and is best tackled by an experimental research program that is specifically designed to address the challenges of mediation analysis.

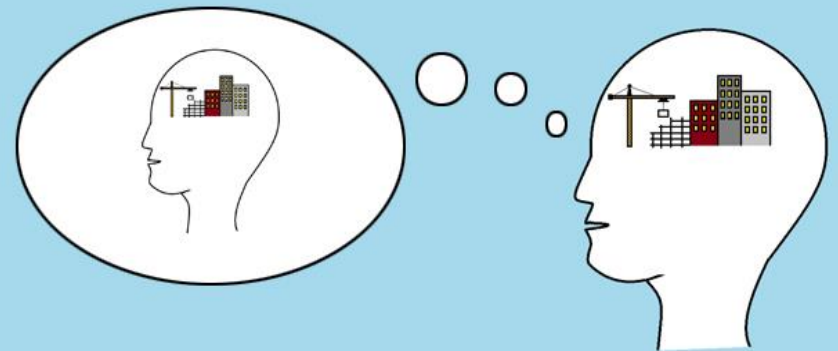
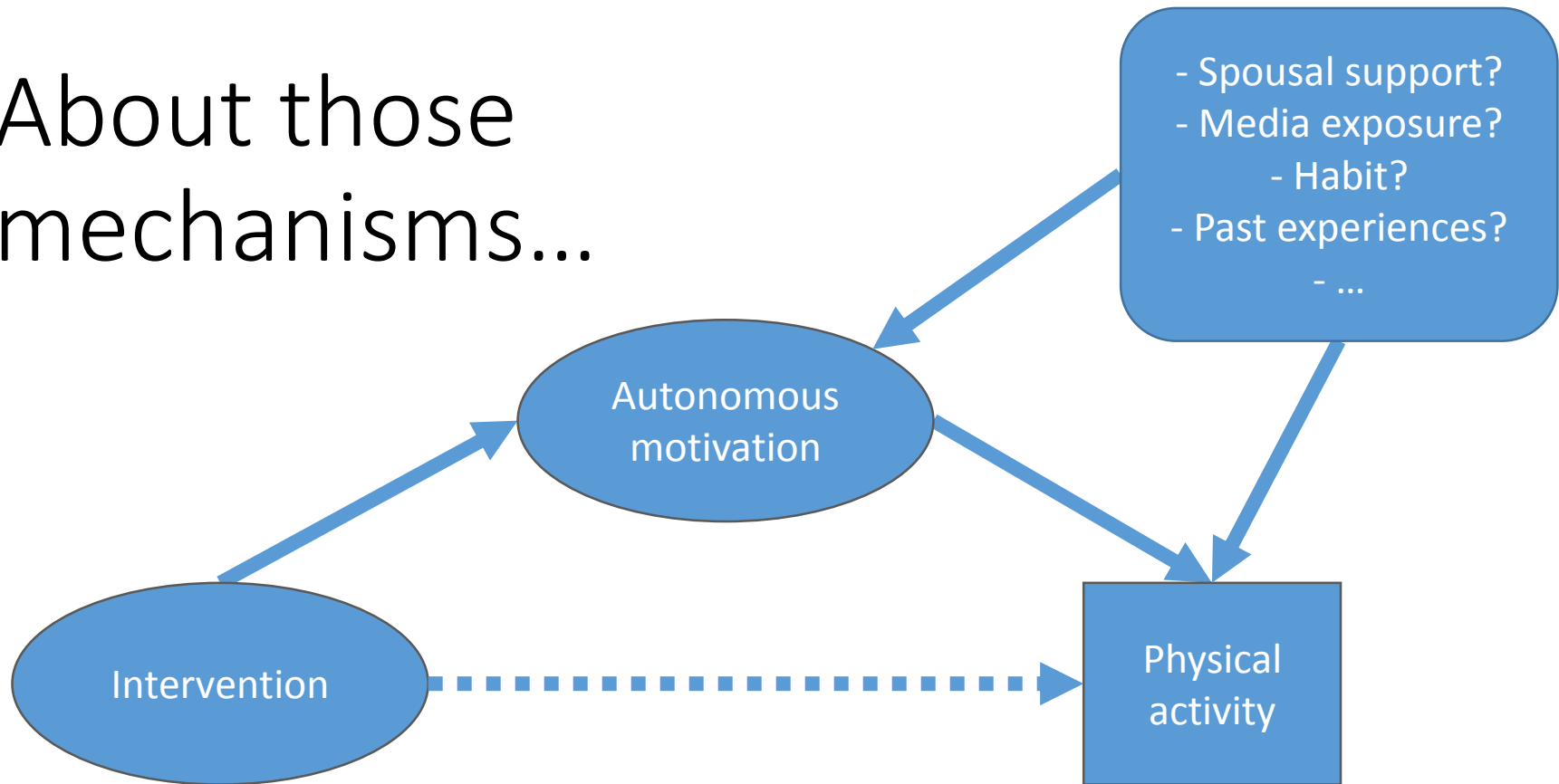


About those mechanisms...

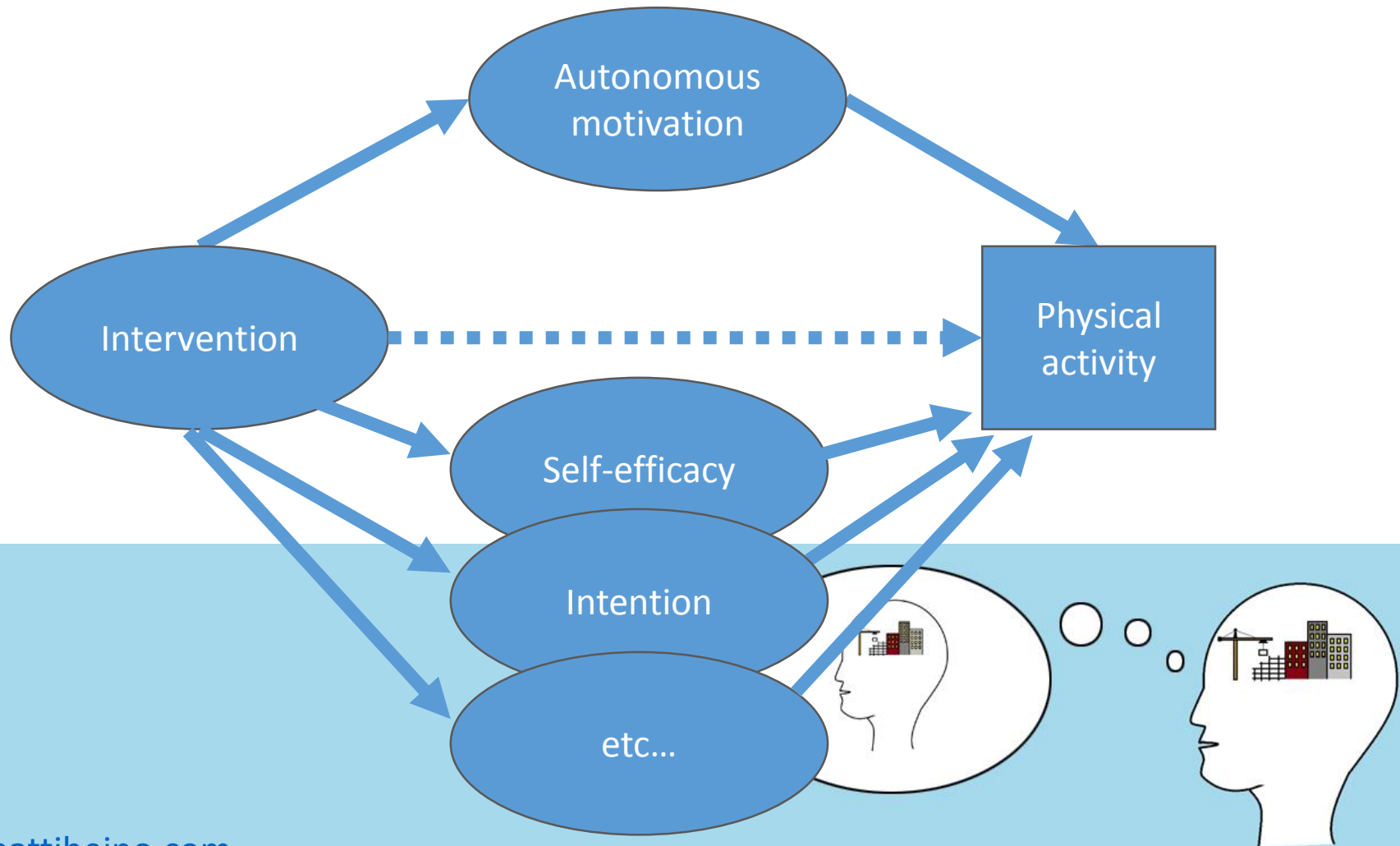


(See [here](#) for an additional sample size problem)

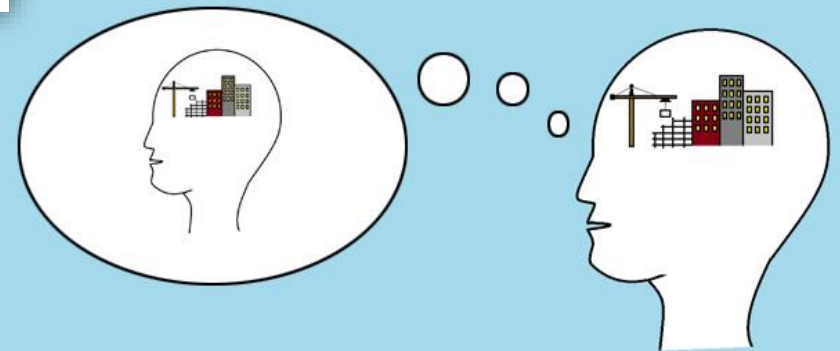
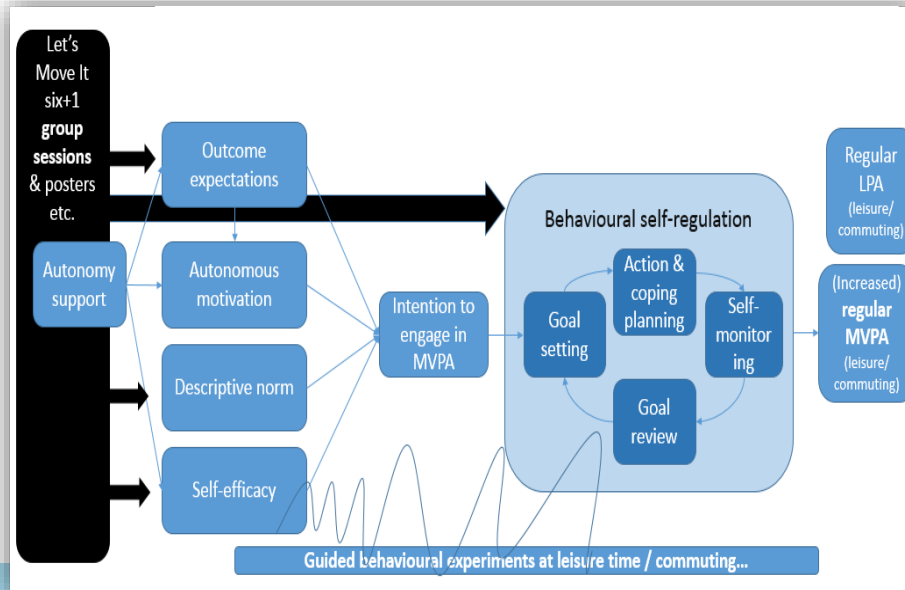
About those mechanisms...



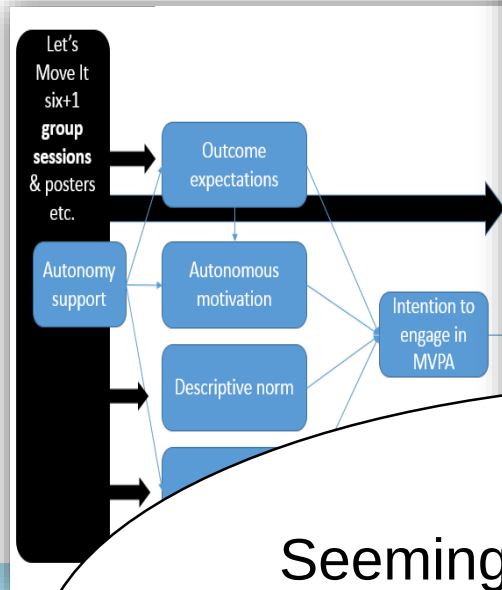
About those mechanisms...



Overfitting



Overfitting



Choosing prediction over explanation in psychology:

Lessons from machine learning

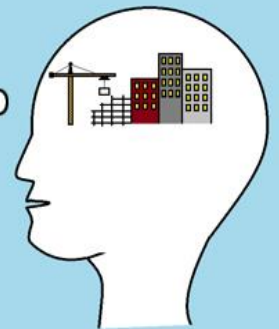
Tal Yarkoni*

Jacob Westfall

at Austin

(2016) [Link](#)

Seemingly supported theories **fail to generalise** to slightly different situations, or other samples from the same population!



April 2017

Evaluation of Evidence of Statistical Support and Corroboration of Subgroup Claims in Randomized Clinical Trials

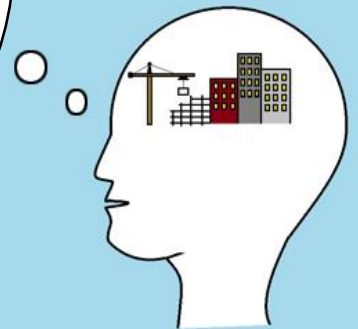
Joshua D. Wallach, MS, PhD^{1,2}; Patrick G. Sullivan, PhD³; et al

» Author Affiliations

JAMA Intern Med. 2017;177(4):341-350

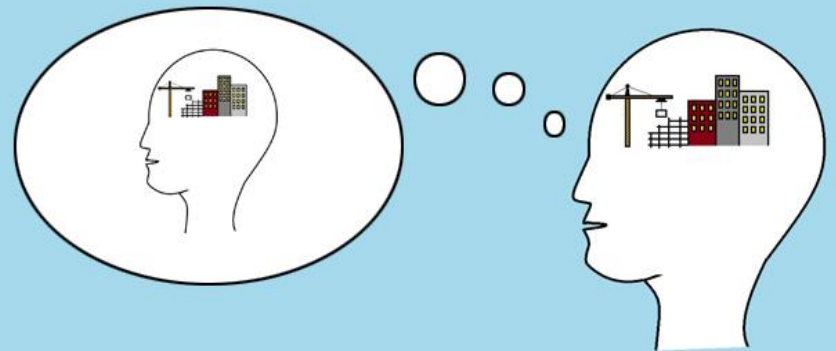
[Link](#)

“Attempts to corroborate statistically significant subgroup differences are rare; when done, the initially **observed** subgroup differences are not reproduced.”



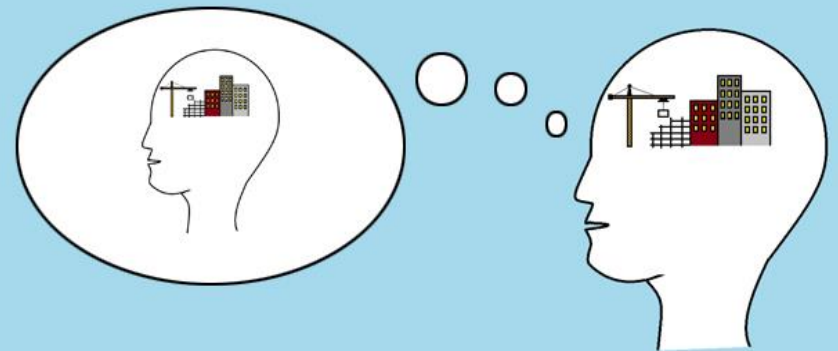
Thanks! Summary and open questions:

- Scientific knowledge grows by risky tests of theories; replication and falsification. Currently, we do neither.
 - Are program theories inherently different to other scientific theories?



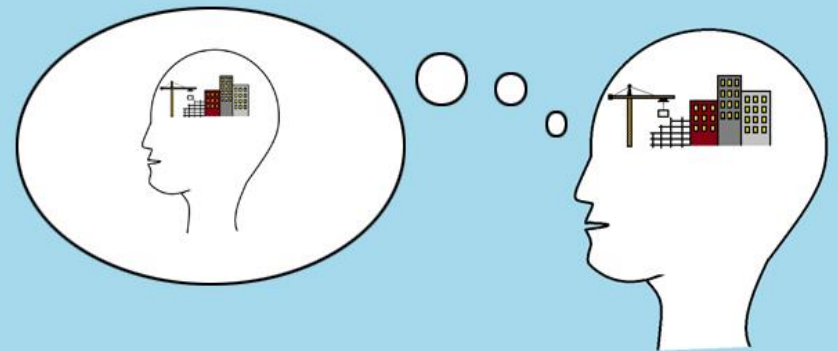
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- Mechanisms are extremely difficult to study. Currently, we claim mediation but remain ignorant about causation.
 - (How) should we study mechanisms of action in complex interventions for complex systems?



Thanks! Summary and open questions:

- Scientific knowledge grows by risky tests of theories; replication and falsification. Currently, we do neither.
 - Are program theories inherently different to other scientific theories?
- Mechanisms are extremely difficult to study. Currently, we claim mediation but remain ignorant about causation.
 - (How) should we study mechanisms of action in complex interventions for complex systems?
- Overfitting is a problem, though cross-validation can help. Currently, sample sizes remain inadequate.
 - How to make models simple enough, but not simpler?



Bonus slide

I'd be happy to have a conversation on how all this applies to real life decision making!

3

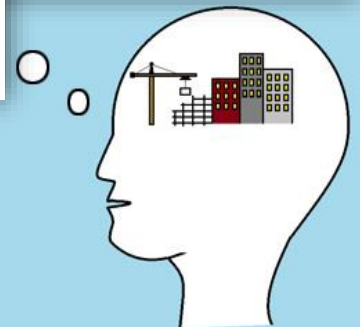
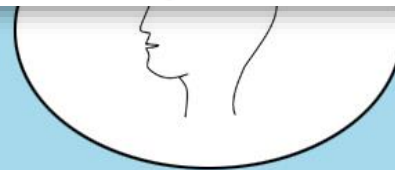
When Simple Is Hard to Accept

Robin M. Hogarth*

In a world in which information is relatively scarce, and where problems for decision are few and simple, information is almost always a positive good. In a world where attention is a major scarce resource, information may be an expensive luxury, for it may turn our attention from what is important to what is unimportant. We cannot afford to attend to information simply because it is there.

Herbert Simon

In summary, if—in several professional domains—human judgments using all available information were replaced by statistical models using only a few variables, the accuracy of predictions could be increased significantly. Perhaps the major obstacle to



Bonus slide 2

How do we figure out,
which pieces of
information truly count,
so that we can ditch the
misleading ones?

The Limits of Phenomenology: From Behaviorism to Drug Testing and Engineering Design

[Link](#)

New England Complex Sy

From Big Data To Important Information

Yaneer Bar-Yam

[Link](#)

in detail. If we want to say anything meaningful about a system—meaningful in the sense of scientific replicability or in terms of utility of knowledge—the only description that is important is one that has universality, i.e. is independent of details. **There is no utility to information that is only true in a particular instance.** Thus, all of scientific inquiry should be understood as an inquiry into universality—the determination of the degree to which information is general or specific.

Recognizing that real world conditions cannot be replicated, our ability to identify important factors that influence outcomes is essential. Ultimately, the possibility of knowledge itself must rely on the ability to differentiate between different pieces of information based on their importance. **Observations must focus on those pieces of information and not on the rest.** Without such an ability, the listing of always irrelevant facts is unavoidable. Thus,

